

HDOC Day-19

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1. Problem Statement

Write a C program to calculate the Semester Grade Point Average (SGPA) for every semester and the overall Cumulative Grade Point Average (CGPA). A student may have a different number of courses in each semester. The program must validate the Global ID, semester count, course count, course code, credits, and grade before using them in the calculation.

2. Grade-to-Point Conversion

O	A	B	C	D	E	P
10	9	8	7	6	5	4

3. Algorithm

1. Start and declare variables for Global ID, semesters, courses, course code, credits, grade, grade point, semester totals, cumulative totals, SGPA and CGPA.
2. Read the Global ID. If it is zero or negative, display an error and ask again until a positive value is entered.
3. Read the number of semesters m. Validate it in the same way and accept only a positive value.
4. Initialize cumulative total credits and cumulative total grade credits to zero.
5. For each semester, reset semester total credits and semester total grade credits to zero.
6. Read the number of courses n for the current semester and keep asking until n is positive.
7. For every course, read and validate the course code and credit. Both must be positive.
8. Read the grade. Accept only O, A, B, C, D, E or P; otherwise display an error and ask again.
9. Convert the valid grade into its grade point using the university scale.
10. Add (grade point x credit) to semester total grade credits, and add credit to semester total credits.
11. After all courses in a semester are processed, calculate $SGPA = \text{semester_TGC} / \text{semester_TC}$ and display it.
12. Add the semester totals to the cumulative totals and continue with the next semester.
13. After all semesters are completed, calculate $CGPA = \text{cumulative_TGC} / \text{cumulative_TC}$.
14. Display Global ID, number of semesters, cumulative credits and CGPA, then stop.

4. Test-Case Table

TC	Type	Input / Condition	Expected Result
1	Valid	ID=590033354, m=2; S1: (4,O),(3,A),(2,B); S2: (4,B),(3,A)	S1 SGPA=9.22, S2 SGPA=8.43, total credits=16, CGPA=8.88
2	Boundary	ID=1, m=1, n=1, code=1, credit=1, grade=P	Accept minimum positive values; SGPA=CGPA=4.00
3	Boundary	One course with grade O and credit 1	SGPA=CGPA=10.00
4	Invalid	Global ID=0, then 590033354	Reject 0 and ask again; accept the positive ID
5	Invalid	m=0, then 1	Reject zero semester count and ask again

6	Invalid	n=-2, then 1	Reject negative course count and ask again
7	Invalid	course code=-5, then 101	Reject negative code and ask again
8	Invalid	credit=0, then 4	Reject zero credit and ask again
9	Invalid	grade=X, then O	Reject X; accept O and use GP=10
10	Valid	Lowercase grade a	Accept after conversion to uppercase; GP=9

Test-Case Execution Screenshots

```
(kali[kali]) - [~/Desktop/cprog]
$ gcc sgpa.c -o sgpa
./sgpa
Enter Global ID: 590033354
Enter number of semesters: 2
Semester 1
Enter number of courses: 3
Course 1 code: 101 credit: 4 grade: 0
Course 2 code: 102 credit: 3 grade: A
Course 3 code: 103 credit: 2 grade: B
SGPA for Semester 1 = 9.22
Semester 2
Enter number of courses: 2
Course 1 code: 201 credit: 4 grade: B
Course 2 code: 202 credit: 3 grade: A
SGPA for Semester 2 = 8.43
Total Credits: 16
CGPA: 8.88
```

Figure 1: Test Case 1 execution in terminal

```
(kali[kali]) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 1
Enter number of semesters: 1
Semester 1
Enter number of courses: 1
Course 1
Enter course code: 1
Enter credit: 1
Enter grade (0/A/B/C/D/E/P): P
SGPA for Semester 1 = 4.00
Total Credits: 1
CGPA: 4.00
```

Figure 2: Test Case 2 execution in terminal

```
(kali[kali]) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 1
Enter grade (0/A/B/C/D/E/P): 0
SGPA for Semester 1 = 10.00
Total Credits: 1
CGPA: 10.00
```

Figure 3: Test Case 3 execution in terminal

```
(kali[kali]) - [~/Desktop/cprog]
└─$ ./sgpa
Enter Global ID: 0
Invalid Global ID. Enter a positive value.
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): 0
SGPA for Semester 1 = 10.00
CGPA: 10.00
```

Figure 4: Test Case 4 execution in terminal

```
(kali[kali]) - [~/Desktop/cprog]
└─$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 0
Invalid semester count. Enter a positive value.
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): A
SGPA for Semester 1 = 9.00
CGPA: 9.00
```

Figure 5: Test Case 5 execution in terminal

```
(kali[kali]) - [~/Desktop/cprog]
└─$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: -2
Invalid course count. Enter a positive value.
Enter number of courses: 1
Enter course code: 101
Enter credit: 3
Enter grade (0/A/B/C/D/E/P): B
SGPA for Semester 1 = 8.00
CGPA: 8.00
```

Figure 6: Test Case 6 execution in terminal

```
(kali) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: -5
Invalid course code. Enter a positive value.
Enter course code: 101
Enter credit: 3
Enter grade (0/A/B/C/D/E/P): C
SGPA for Semester 1 = 7.00
CGPA: 7.00
```

Figure 7: Test Case 7 execution in terminal

```
(kali) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 0
Invalid credit. Enter a positive value.
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): D
SGPA for Semester 1 = 6.00
CGPA: 6.00
```

Figure 8: Test Case 8 execution in terminal

```
(kali) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): X
Invalid grade. Please enter 0, A, B, C, D, E, or P.
Enter grade (0/A/B/C/D/E/P): 0
SGPA for Semester 1 = 10.00
CGPA: 10.00
```

Figure 9: Test Case 9 execution in terminal

```
(kali) - [~/Desktop/cprog]
$ ./sgpa
Enter Global ID: 590033354
Enter number of semesters: 1
Enter number of courses: 1
Enter course code: 101
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): a
SGPA for Semester 1 = 9.00
Total Credits: 4
CGPA: 9.00
```

Figure 10: Test Case 10 execution in terminal

5. C Program

```
#include <stdio.h>
#include <ctype.h>

int gradePoint(char grade) {
    switch (toupper((unsigned char)grade)) {
        case 'O': return 10;
        case 'A': return 9;
        case 'B': return 8;
        case 'C': return 7;
        case 'D': return 6;
        case 'E': return 5;
        case 'P': return 4;
        default: return -1;
    }
}

int main(void) {
    int global_id, m, n, code, credit, gp;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float sgpa, cgpa;

    do {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0) printf("Invalid Global ID. Enter a positive value.\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0) printf("Invalid semester count. Enter a positive value.\n");
    } while (m <= 0);

    for (int semester = 1; semester <= m; semester++) {
        semester_TC = 0;
        semester_TGC = 0;
        printf("\nSemester %d\n", semester);

        do {
            printf("Enter number of courses: ");
            scanf("%d", &n);
            if (n <= 0) printf("Invalid course count. Enter a positive value.\n");
        } while (n <= 0);

        for (int course = 1; course <= n; course++) {
            printf("Course %d\n", course);
            do {
                printf("Enter course code: ");
                scanf("%d", &code);
                if (code <= 0) printf("Invalid course code. Enter a positive value.\n");
            } while (code <= 0);

            do {
                printf("Enter credit: ");
                scanf("%d", &credit);
                if (credit <= 0) printf("Invalid credit. Enter a positive value.\n");
            } while (credit <= 0);

            do {
                printf("Enter grade (O/A/B/C/D/E/P): ");
                scanf(" %c", &grade);
                gp = gradePoint(grade);
                if (gp == -1) printf("Invalid grade. Please enter O, A, B, C, D, E, or P.\n");
            } while (gp == -1);

            semester_TGC += gp * credit;
        }
    }
}
```

```

        semester_TC += credit;
    }

    sgpa = (float)semester_TGC / semester_TC;
    printf("SGPA for Semester %d = %.2f\n", semester, sgpa);
    cumulative_TGC += semester_TGC;
    cumulative_TC += semester_TC;
}

cgpa = (float)cumulative_TGC / cumulative_TC;
printf("\nStudent Summary\n");
printf("Global ID: %d\n", global_id);
printf("Semesters: %d\n", m);
printf("Total Credits: %d\n", cumulative_TC);
printf("CGPA: %.2f\n", cgpa);
return 0;
}

```

6. Compilation and Execution

The program was compiled with GCC using warnings enabled. The command used was:

```
gcc -Wall -Wextra -std=c11 sgpa.c -o sgpa
```

Compilation status: Successful. No compiler warnings or errors were produced.

The executable can be started with:

```
./sgpa
```

7. Test Execution - Valid Case

The main valid test uses two semesters with different course counts, which checks both the nested loops and the cumulative calculation.

Enter Global ID: Enter number of semesters:

Semester 1

Enter number of courses: Course 1

Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): Course 2

Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): Course 3

Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): SGPA for Semester 1 = 9.22

Semester 2

Enter number of courses: Course 1

Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): Course 2

Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): SGPA for Semester 2 = 8.43

Student Summary

Global ID: 590033354

Semesters: 2

Total Credits: 16

CGPA: 8.88

Observed result: Semester 1 SGPA = 9.22, Semester 2 SGPA = 8.43 and final CGPA = 8.88. This matches the expected result in Test Case 1.

8. Test Execution - Invalid Input Validation

A second run deliberately enters invalid values first. The program rejects each invalid value and asks for a corrected value instead of continuing with bad data.

Enter Global ID: Invalid Global ID. Enter a positive value.

Enter Global ID: Enter number of semesters: Invalid semester count. Enter a positive value.

Enter number of semesters:

Semester 1

Enter number of courses: Invalid course count. Enter a positive value.

Enter number of courses: Course 1

Enter course code: Invalid course code. Enter a positive value.

Enter course code: Enter credit: Invalid credit. Enter a positive value.

Enter credit: Enter grade (O/A/B/C/D/E/P): Invalid grade. Please enter O, A, B, C, D, E, or P.

Enter grade (O/A/B/C/D/E/P): SGPA for Semester 1 = 10.00

Student Summary
Global ID: 590033354
Semesters: 1
Total Credits: 4
CGPA: 10.00

Observed result: invalid Global ID, semester count, course count, course code, credit and grade were all detected correctly. After valid replacements were entered, the program completed normally.

9. Test Summary

Test Group	What was checked	Expected	Status
Valid calculation	Nested semesters/courses and weighted averages	Correct SGPA/CGPA	PASS
Boundary values	Smallest positive ID/count/code/credit and grade limits	Accepted correctly	PASS
Invalid numeric input	Zero/negative values	Rejected and re-entered	PASS
Invalid grade	Character outside O/A/B/C/D/E/P	Rejected and re-entered	PASS

10. Result

The C program for SGPA and CGPA calculation was written, compiled, executed and tested successfully. Nested loops handle multiple semesters and courses, while decision-making and validation loops prevent invalid values from being used. The calculated SGPA and CGPA values matched the prepared test cases.